

Analysis of Broadband Psi (Ψ)-Shaped Microstrip Antennas

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Abstract

A new Ψ -shaped microstrip antenna is reported, which increased the bandwidth of the E-shaped microstrip antenna by cutting an additional pair of slots on the other radiating edge of the E-shaped patch. The Ψ -shaped patch yielded a bandwidth of nearly 60% at a center frequency of around 5500 MHz. It gave a maximum gain of more than 10 dBi, which reduced to less than 4 dBi towards the higher frequencies of the bandwidth. In this paper, the broadband responses of the E-shaped and the reported Ψ -shaped patches are studied. In the E-shaped patch, the pair of rectangular slots did not introduce any mode, but modified the TM_{20} mode's resonance frequency of the patch and, along with the fundamental TM_{01} mode, resulted in the broadband response. Furthermore, when an additional pair of slots were cut to realize the Ψ -shaped patch, they modified the higher-order TM_{21} mode's resonance frequency of the patch. Along with the modified TM_{20} and TM_{01} modes, this yielded a larger bandwidth. Since the TM_{21} mode was present towards the higher frequencies of the bandwidth, the gain was reduced to less than 4 dBi. Furthermore, a proximity-fed design of a Ψ -shaped microstrip antenna in the same frequency band was proposed. The proposed configuration gave a higher bandwidth compared to the reported Ψ -shaped patch, with better gain characteristics over the bandwidth. A proximity-fed design of the Ψ -shaped patch in the 1000 MHz frequency band was also proposed. It gave a bandwidth of more than 50% with a broadside radiation pattern, and a gain of more than 8 dBi over the complete bandwidth.

Keywords: Broadband rectangular microstrip antenna; E-shaped microstrip antenna; Ψ -shaped microstrip antenna; proximity feed

1. Introduction

The commonly used technique for realizing a broadband microstrip antenna (MSA) is to cut slots of different shapes – such as a U-shaped slot, a V-shaped slot, an L-shaped slot, or a pair of rectangular or toothbrush shaped slots – at an appropriate position inside the patch [1-12]. The general understanding of these broadband slot-cut microstrip antennas is that when the slot length equals either a half-wavelength or a quarter-wavelength, then the slot introduces a mode near the fundamental-mode resonance frequency of the patch, and yields a broader bandwidth (BW). A compact microstrip antenna is

realized by placing the shorting posts along the zero-field line, or by using the symmetry of the patch configuration across the feed-point axis at its fundamental mode [2]. Compact and broadband slot-cut microstrip antennas are obtained by using the symmetry of these slot-cut broadband microstrip antennas across the feed-point axis [11-14]. The resulting configuration operates in the same frequency band, but with half the patch size. A further increase in the bandwidths of slot-cut compact and broadband microstrip antennas is realized by cutting more than one slot inside or on the edges of the patch [15-19]. These broadband slot-cut microstrip antennas are optimized on thicker substrates, with $h \geq 0.06\lambda_0$ to $0.08\lambda_0$. However, for a substrate

thickness greater than $0.08\lambda_0$, the bandwidths of these antennas are limited by feed-probe inductance. Using a simpler proximity feeding technique, the bandwidths of slot-cut microstrip antennas are increased for a substrate thickness greater than $0.08\lambda_0$ [20, 21].

To realize broadband responses in the desired frequency band in these slot-cut microstrip antennas, the slot length is taken to be either equal to a half-wavelength or a quarter-wavelength in length, depending on the position of the slot in the patch. However, it was observed that the surface currents around the shorted end of the slot encircle a finite length, which adds to the additional length. Hence, while calculating the slot length, an additional correction length is added to the quarter- or half-wavelength approximation [22]. However, this simpler approach of calculating the slot length does not give accurate results for different lengths and different positions inside the patch. An analysis of the effects of a U-shaped slot on the broadband and dual-band responses of a rectangular microstrip antenna (RMSA) was reported in [23]. The equations for the resonance frequencies at different modes of the U-shaped slot-cut rectangular microstrip antennas were given. However, plots of the frequencies calculated using the proposed equations as functions of the measured and simulated data were not given. A clear description of the various modes at different frequencies was also not discussed. By studying the resonance-curve plots, surface-current distributions, and the radiation patterns for different slot lengths and their positions, the dual-band response of the slot-cut circular microstrip antenna was studied [24]. It was observed that the slot modified the higher-order mode of the patch and, along with the fundamental mode, realized dual frequencies. A broadband E-shaped microstrip antenna is realized by cutting a pair of rectangular slots on one of the radiating edges of a rectangular microstrip antenna [10]. Recently, the bandwidth of the E-shaped microstrip antenna was increased by cutting an additional pair of slots on the other radiating edges of the E-shaped patch, resulting in a Ψ -shaped microstrip antenna [19]. The Ψ -shaped microstrip antenna gave a bandwidth of 60% (3.82 GHz to 7.10 GHz) on an infinite ground plane, and a bandwidth of 55% (4.15 GHz to 7.30 GHz) on a finite ground plane, with a broadside radiation pattern and a peak gain of more than 10 dBi [19]. However, towards the higher frequencies of the bandwidth, the gain reduced to less than 4 dBi. The broader bandwidth in the Ψ -shaped microstrip antenna was reported to be due to the better control of the current distributions towards the higher frequencies of the bandwidth, which was realized due to the tuning stub that was formed by cutting an additional pair of rectangular slots [19]. However, which additional mode led to the increase in bandwidth and a decrease in gain to as low as 3 dBi was not explained.

In this paper, a detailed analysis of the broadband behavior of the Ψ -shaped microstrip antenna is presented. First, the equivalent rectangular microstrip antenna and the E-shaped patch were studied. The response of the E-shaped microstrip antenna for different slot lengths was analyzed by studying its resonance-curve plots, surface-current distributions, and the simulated radiation-pattern plots. They were compared with the

plots of the equivalent rectangular microstrip antenna. It was observed that the pair of rectangular slots that produces the E-shaped structure did not introduce any mode, but modified the orthogonal TM_{20} mode frequency of the rectangular microstrip antenna and, along with the TM_{01} mode, produced the broadband response. The radiation pattern of the TM_{01} mode was in the broadside direction, with the E and H planes aligned along $\Phi = 90^\circ$ and 0° , respectively. Conventionally, the pattern of the TM_{20} mode is conical (i.e., maximum along the end-fire direction), with the E and H planes aligned along $\Phi = 0^\circ$ and 90° , respectively. Since the pair of slots modified the directions of the surface currents of the TM_{20} mode, the radiation pattern over the bandwidth in the E-shaped microstrip antenna remained in the broadside direction. When the additional pair of slots was cut inside the E-shaped microstrip antenna to realize the Ψ -shaped microstrip antenna, these slots modified the higher-order TM_{21} -mode resonance frequency of the equivalent rectangular microstrip antenna. Due to their coupling with the modes of the E-shaped patch (i.e., the TM_{01} and modified TM_{20} modes), they realized a broader bandwidth. The radiation pattern of the TM_{21} mode was conical, with higher cross-polar levels. The pair of slots modified the directions of the surface currents of the TM_{21} mode. Since this mode was dominant towards the higher frequencies of the bandwidth, the radiation patterns at those frequencies showed higher cross-polarization levels, and the antenna gain reduced from 10 dBi to as low as 3 dBi.

A proximity-fed design of the Ψ -shaped microstrip antenna in the same frequency band is proposed. The proximity fed Ψ -shaped microstrip antenna gave a bandwidth of more than 4000 MHz (>65%). Compared to the reported probe-fed Ψ -shaped microstrip antenna, the proximity-fed microstrip antenna showed a gain variation of only 2.5 dBi (i.e., from 9.5 dBi to 7 dBi) towards the higher frequencies of the bandwidth.

A broadband proximity-fed design of the Ψ -shaped microstrip antenna in the 1000 MHz frequency band is also proposed. This configuration gave a measured bandwidth of 524 MHz (47.7%), with a peak gain very close to 10 dBi. In this frequency band, due to the use of the proximity feed, the antenna gain also reduced towards higher frequencies to only 7 dBi, from its peak value of 10 dBi.

The proposed microstrip antennas were first analyzed using the *IE3D* software, followed by experimental verifications [25]. An air substrate was used to maximize the radiation efficiency. The experiment was carried out using an HP vector network analyzer. In the *IE3D* simulations, an infinite ground plane was used. To simulate this effect in the 5500 MHz frequency band, a square ground plane with a side length of 15 cm ($2.7\lambda_0$) was used, whereas for the 1000 MHz frequency band, a square ground plane with a side length of 80 cm ($2.7\lambda_0$) was used. Furthermore, in the 1000 MHz frequency band, the antenna's response was also measured using a finite square ground plane of side length 28 cm ($0.93\lambda_0$). In the 5500 MHz

band, the antenna was fed using an SMA connector with an inner wire diameter of 0.12 cm, whereas in the 1000 MHz band, the antenna was fed using a type-N connector with an inner wire diameter of 0.32 cm. The radiation patterns were measured in minimum-reflection surroundings, with the required minimum far-field distance between the reference antenna and the antenna under test at all the frequencies [26]. The gain was measured using the three-antenna method [26].

2. The Broadband Ψ -Shaped Microstrip Antenna

The broadband Ψ -shaped microstrip antenna is shown in Figures 1a and 1b. The dimensions shown in all the figures are in cm. The patch was fabricated on a microwave substrate ($\epsilon_r = 2.50$, $h = 0.33$ cm, $\tan \delta = 0.002$), and it was suspended above the ground plane using a foam substrate ($\epsilon_r = 1.06$, $h = 0.6$ cm, $\tan \delta = 0.0002$). The Ψ -shaped microstrip antenna on an infinite ground plane gave a bandwidth from 3820 MHz to 7100 MHz (60%), as shown in Figure 1c [19]. This was more than the bandwidth of an equivalent E-shaped

microstrip antenna [19]. The radiation pattern was in the broadside direction, with the E and H planes aligned along $\Phi = 90^\circ$ and 0° , respectively. However, the antenna showed higher cross-polar radiation towards the higher frequencies of the bandwidth, which reduced its gain from the peak value of 10.5 dBi to less than 4 dBi, as shown in Figure 1d [19]. The increase in bandwidth for the Ψ -shaped microstrip antenna was attributed to the mode introduced by the tuning stub (as shown in Figure 1a), which was formed by cutting an additional pair of rectangular slots on the edges of the E-shaped microstrip antenna [19]. However, a clear description of which mode was introduced by the tuning stub (which also resulted in gain reduction) has not been given. To study this, an analysis of the broadband Ψ -shaped microstrip antenna is presented in the following section.

3. An Analysis of the Broadband Ψ -Shaped Microstrip Antenna

The analysis of Ψ -shaped microstrip antenna was carried out on an infinite ground plane. The Ψ -shaped microstrip antenna considered was optimized at a center frequency of

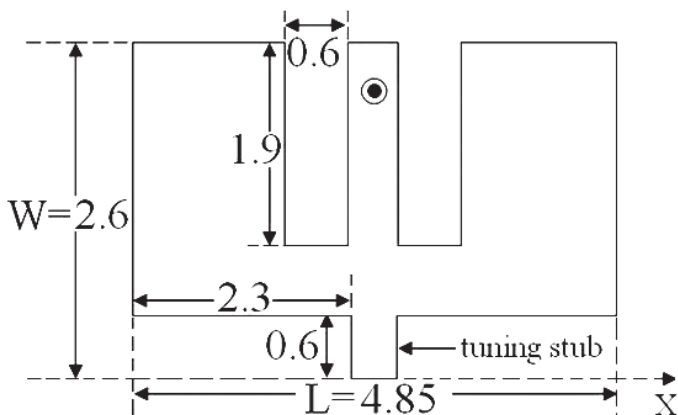


Figure 1a. A top view of the Ψ -shaped microstrip antenna [19].

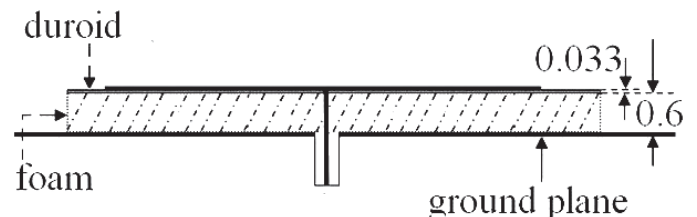


Figure 1b. A side view of the Ψ -shaped microstrip antenna [19].

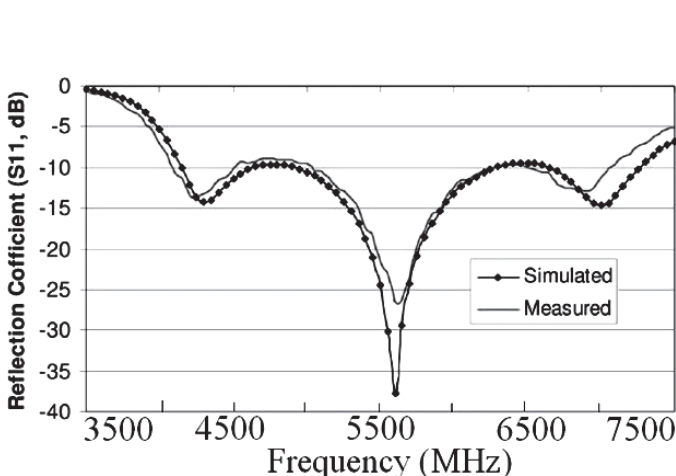


Figure 1c. The return loss over the bandwidth for the Ψ -shaped microstrip antenna [19].

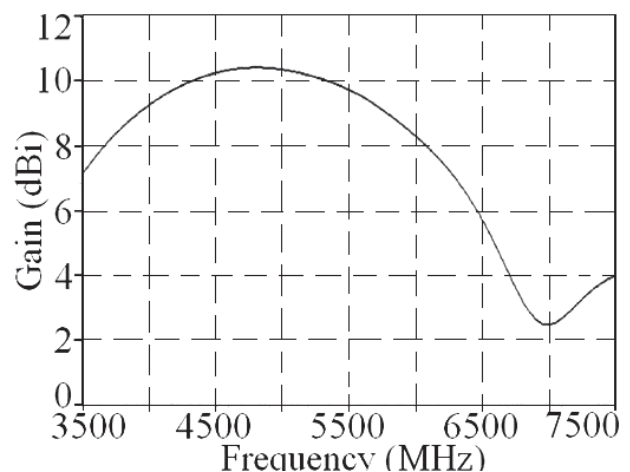


Figure 1d. The gain variation over the bandwidth for the Ψ -shaped microstrip antenna [19].

around 5500 MHz on a total substrate thickness of 0.633 cm ($0.12\lambda_0$). The use of a thicker substrate led to a higher percentage bandwidth for the individual patch modes. The Ψ -shaped microstrip antenna was analyzed using the *IE3D* software. Its simulated input impedance and resonance-curve plots are shown in Figures 2a and 2b. Two loops were observed in the input-impedance plot, whereas three peaks were present in the resonance-curve plot, which indicated the presence of three resonant modes. The second loop was formed near a frequency of around 6500 MHz, at which the third peak in the resonance curve was present. The surface-current distributions at three frequencies over the bandwidth are shown in Figures 2c to 2e. At the second frequency, the currents were directed along the patch's width. At the first and third frequencies, the surface currents were directed along the patch's length as well as its width. The surface currents that were directed along the patch's length reduced the antenna's gain towards the lower and higher frequencies of the bandwidth. To understand this current distribution and the different resonant modes involved in the broadband response, analyses of the equivalent rectangular microstrip antenna, and then further of the E-shaped microstrip antenna, were carried out.

The dimensions of the equivalent rectangular microstrip antenna in the proposed Ψ -shaped microstrip antenna were $L = 4.85$ cm and $W = 2.6$ cm, which gave an L/W ratio of 1.865. This higher ratio resulted in higher gain, but it led to closely spaced orthogonal frequencies. The resonance frequencies (f_{mn}) of the different resonant TM_{mn} modes for this rectangular microstrip antenna were calculated by using Equation (1):

$$f_{mn} = \frac{1}{2\sqrt{\epsilon_{re}}} \sqrt{\left(\frac{m}{L_e}\right)^2 + \left(\frac{n}{W_e}\right)^2}. \quad (1)$$

L_e and W_e are the effective patch dimensions, which account for the fringing-field extensions around the patch's edges [2]. The various resonance frequencies calculated using Equation (1) in the 2000 MHz to 8000 MHz frequency range were TM_{10} : 2497 MHz; TM_{01} : 4124 MHz; TM_{20} : 4994 MHz; TM_{02} : 8248 MHz; TM_{21} : 6476 MHz; TM_{12} : 8620 MHz; and TM_{30} : 7491 MHz. This rectangular microstrip antenna was simulated using the *IE3D* software, and its resonance curve is shown in Figure 3a. Since the rectangular microstrip antenna was fed along the line joining the center of the patch and the center of its length, it did not excite the TM_{10} mode. The first peak in the resonance plot was due to the TM_{01} mode at a frequency equal to 3992 MHz. The next two peaks corresponded to the TM_{20} (4616 MHz) and TM_{21} (6200 MHz) modes. For the given feed location, the TM_{30} mode (7460 MHz) was partially excited. These frequencies were closer to the frequencies obtained using Equation (1) for the corresponding modes. The surface-current distributions for the TM_{01} , TM_{20} , and TM_{21} modes are shown in Figures 3b to 3d. For the TM_{01} mode, the surface currents showed one half-wavelength

variations along the patch width, whereas for the TM_{20} mode, the surface currents showed two half-wavelength variations along the patch length. For the TM_{21} mode, the surface current showed two half-wavelength variations along the patch length and one half-wavelength variation along the patch width. The radiation patterns for the TM_{20} and TM_{21} modes were conical, with higher cross-polarization levels, as shown in Figures 4a and 4b.

A pair of slots was cut inside this rectangular microstrip antenna to realize the E-shaped microstrip antenna, as shown in Figure 4c. The slot length (l_e) was taken to be the same as that used in the reported configuration. The slot width (w_e) was increased in steps of 0.5 cm, and the resonance curves for different values of w_e are shown in Figure 5a. When the slot was cut inside the patch, it increased the length of the surface currents for a given mode, and reduced its frequency. In the E-shaped microstrip antenna, the length of the surface currents along the patch length was increased with an increase in w_e , which reduced the TM_{20} mode's frequency. The slot width was parallel to the surface currents of the TM_{01} mode: its frequency therefore nearly remained unchanged. The TM_{21} -mode frequency of the rectangular microstrip antenna depends upon both of the patch's dimensions. w_e modifies the effective patch length. Therefore, for smaller w_e , the TM_{21} -mode frequency was reduced. However, for larger w_e , the slot effectively reduced the patch's width (W), which increased the TM_{21} -mode frequency. The surface-current distributions for different values of w_e for the TM_{20} mode are shown in Figures 5b and 5c. With an increase in w_e , it was observed that the surface currents were reoriented along the patch's width.

The resonance-curve plots for different values of the separation between the pair of slots (y) are shown in Figure 6a. With a variation in y from 0.3 cm to 1.1 cm, the TM_{30} frequency changed, whereas the other mode frequencies remained nearly fixed. Also, the increase in y increased the impedance at the individual modal frequencies. With an increase in the slot width, the TM_{20} frequency thus was reduced. When its spacing with respect to the TM_{01} mode frequency was optimized, a loop was formed inside the $VSWR = 2$ circle in the input-impedance plot, as shown in Figure 6b. The optimized E-shaped microstrip antenna gave a simulated bandwidth of 1334 MHz (31%). As the slot modified the directions of the surface currents for the TM_{20} mode, the radiation pattern over the bandwidth remained in the broadside direction, with the E and H planes aligned along $\Phi = 90^\circ$ and 0° , respectively.

The input-impedance locus for the E-shaped microstrip antenna showed the formation of a second loop in the higher-frequency region. As seen from the resonance-curve plots, the second loop was formed, due to the TM_{21} mode, at a frequency of 6788 MHz. The surface-current distribution and the radiation-pattern plot at this frequency are shown in Figures 7a and 7b. The surface-current distribution was modified due to the pair of

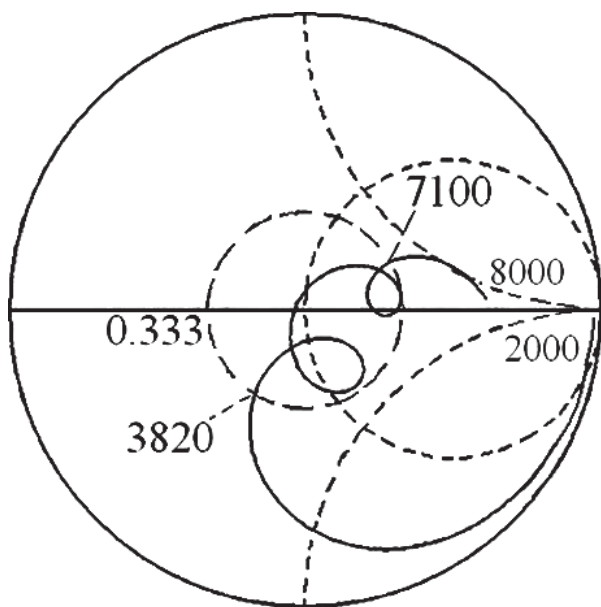


Figure 2a. The simulated input impedance for the Ψ -shaped microstrip antenna over the bandwidth.

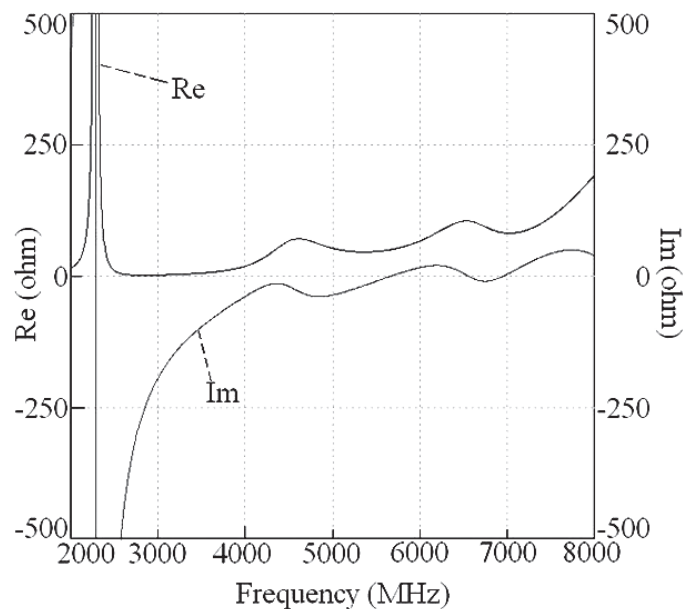
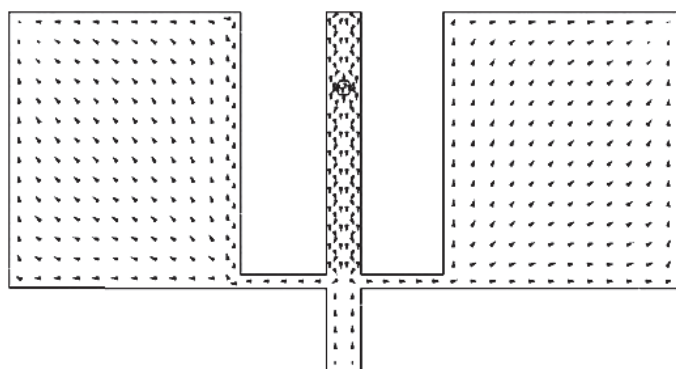
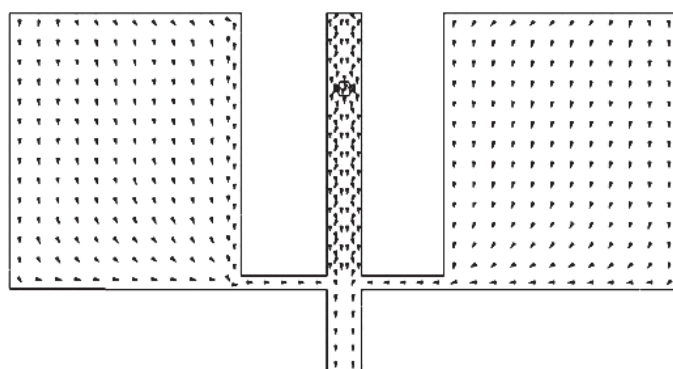


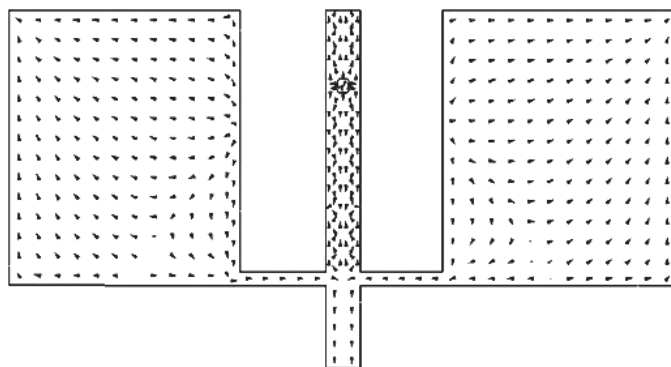
Figure 2b. The simulated resonance curves for the Ψ -shaped microstrip antenna over the bandwidth.



(c) 3800 MHz



(d) 5400 MHz



(e) 7100 MHz

Figures 2c-2e. The simulated surface-current distributions of the Ψ -shaped microstrip antenna over the bandwidth.

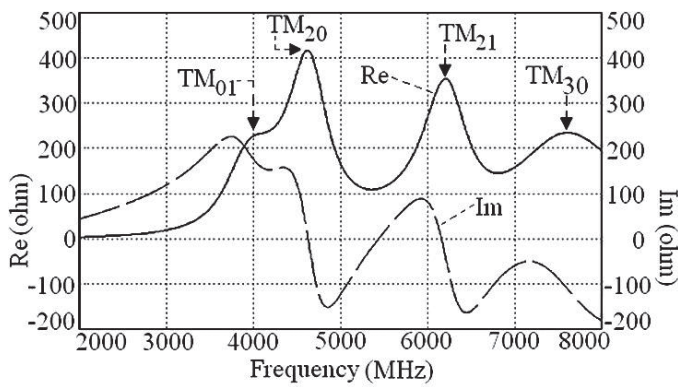
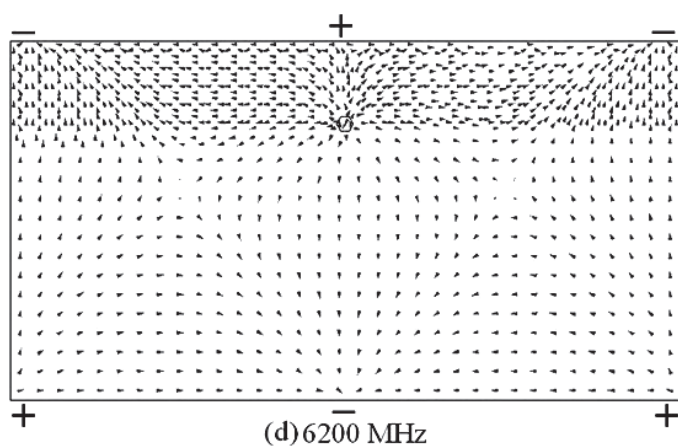
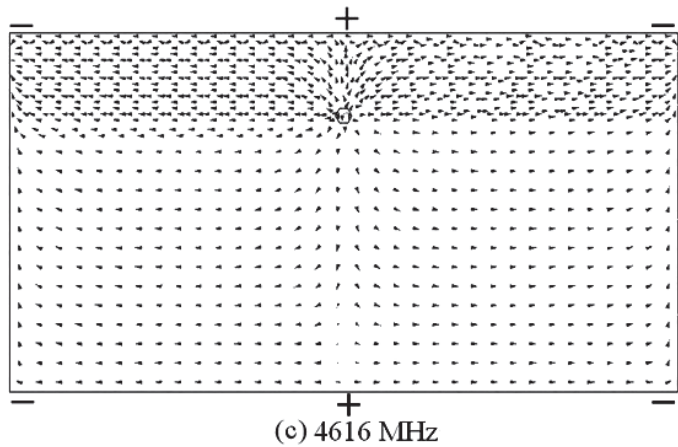
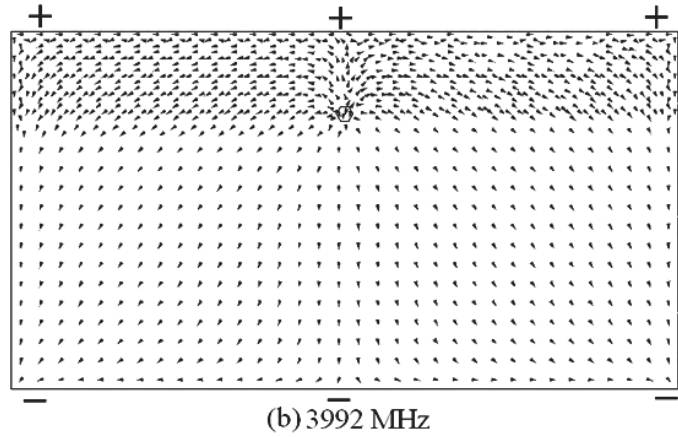
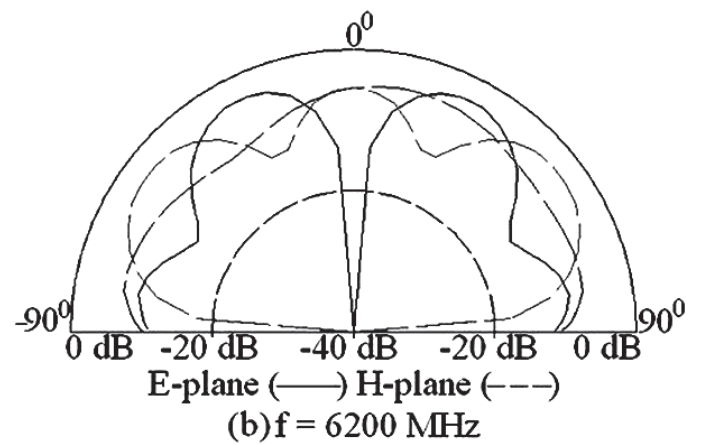
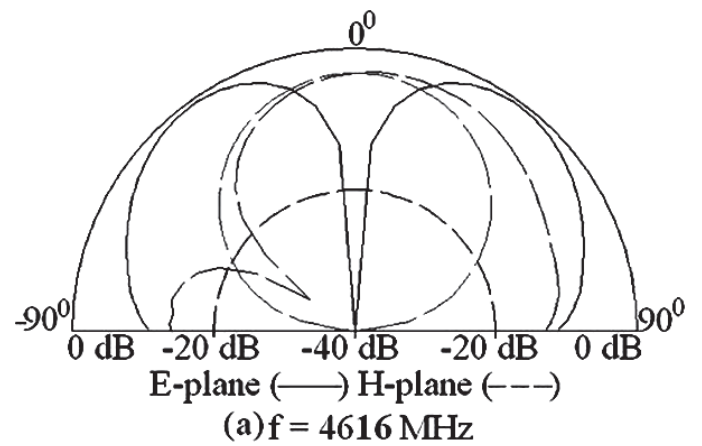


Figure 3a. The resonance curve plot for the equivalent rectangular microstrip antenna.



Figures 3b-3d. The surface-current distributions for the (b) TM_{01} , (c) TM_{20} , and (d) TM_{21} modes for the equivalent rectangular microstrip antenna



Figures 4a, 4b. The radiation patterns for the (a) TM_{20} and (b) TM_{21} modes for the rectangular microstrip antenna.

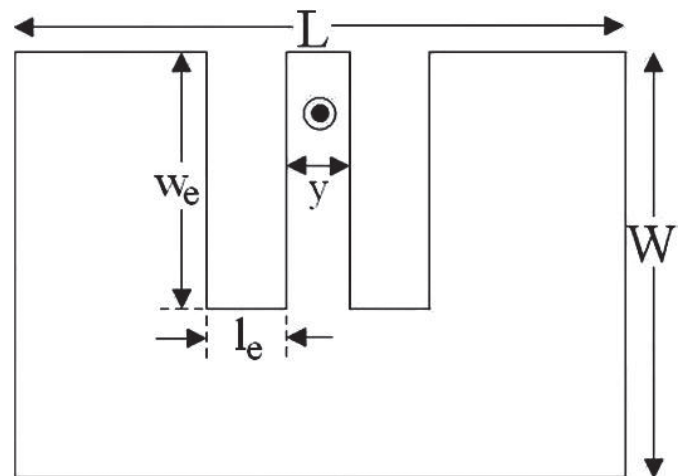


Figure 4c. The E-shaped microstrip antenna.

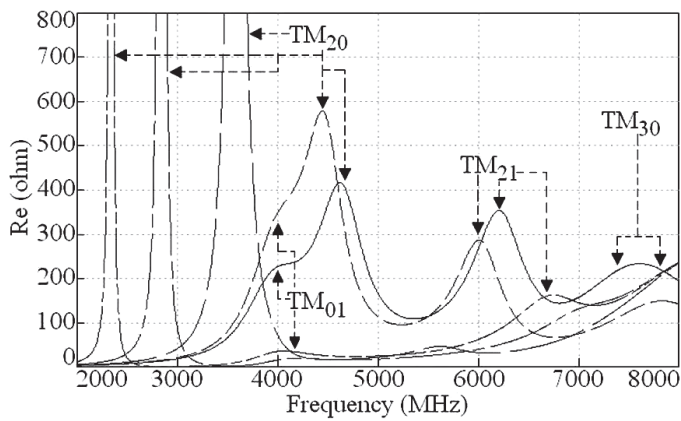


Figure 5a. The resonance curve plots for (a) (—) the rectangular microstrip antenna, and for the E-shaped microstrip antenna for (---) $w_e = 0.5$ cm, (— · —) $w_e = 1.0$ cm, (— — —) $w_e = 1.5$ cm, and (— · · —) $w_e = 1.9$ cm.

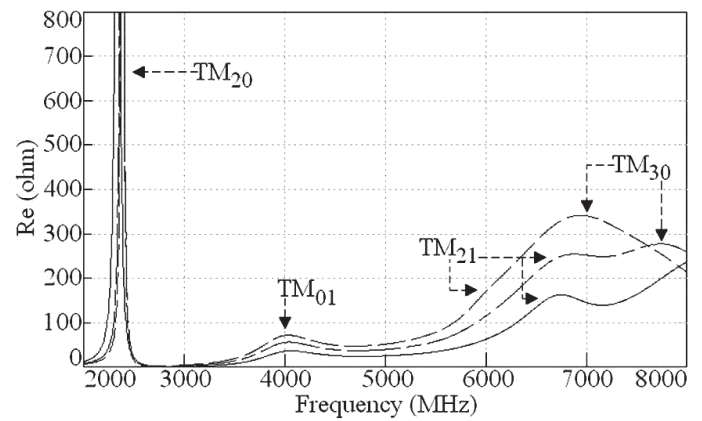
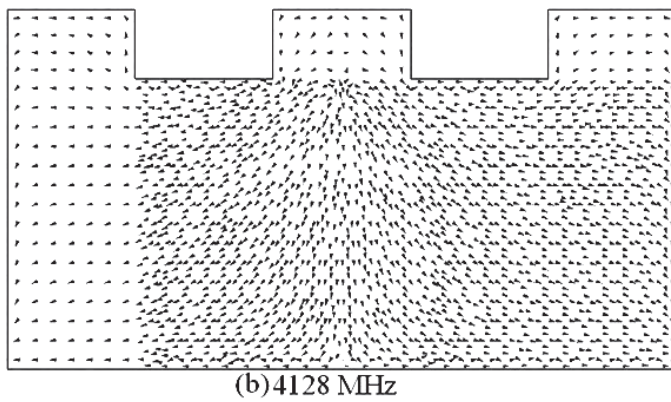
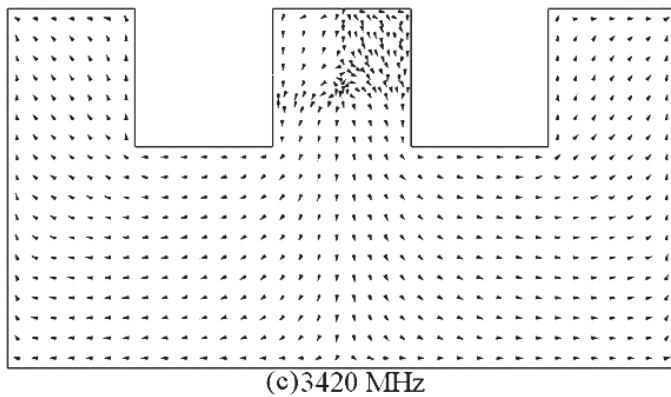


Figure 6a. The resonance curve plot for the E-shaped microstrip antenna for $w_e = 2.0$ cm and (a) (—) $y = 0.3$ cm, (— — —) $y = 0.7$ cm, and (---) $y = 1.1$ cm.



(b) 4128 MHz



(c) 3420 MHz

Figures 5b, 5c. The surface-current distributions for (b) $w_e = 0.5$ cm and (c) $w_e = 1.0$ cm for the E-shaped microstrip antenna.

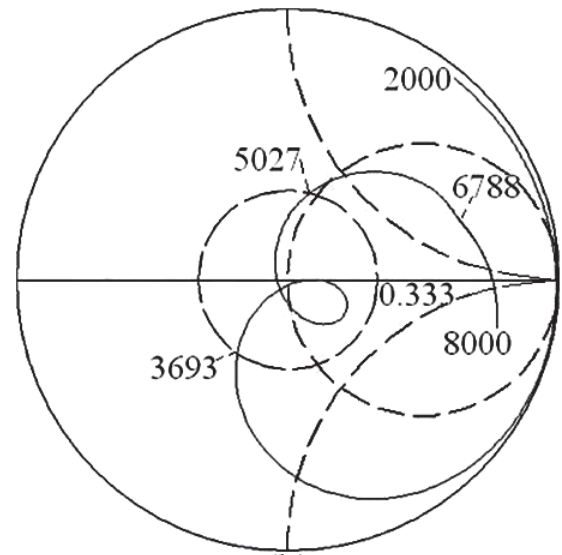


Figure 6b. The optimized input-impedance locus for the E-shaped microstrip antenna.

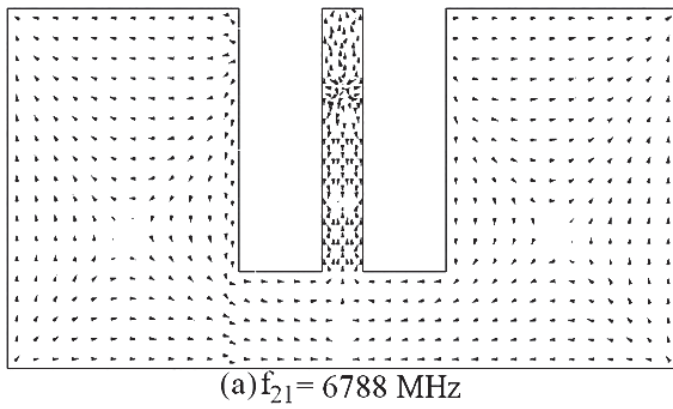


Figure 7a. The surface-current distribution for the TM_{21} mode for the E-shaped microstrip antenna.

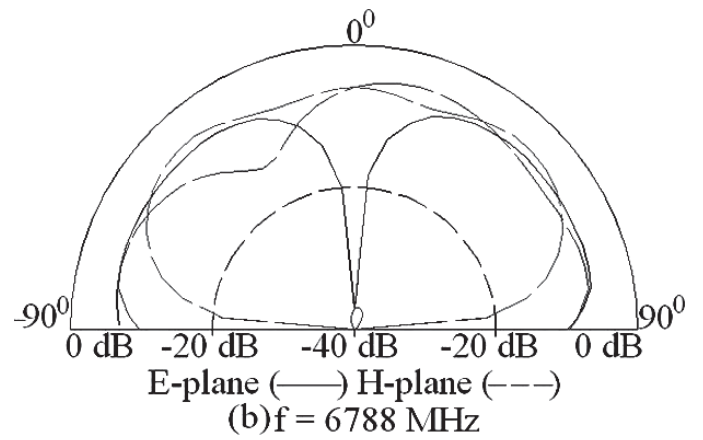


Figure 7b. The radiation pattern for the TM_{21} mode for the E-shaped microstrip antenna.

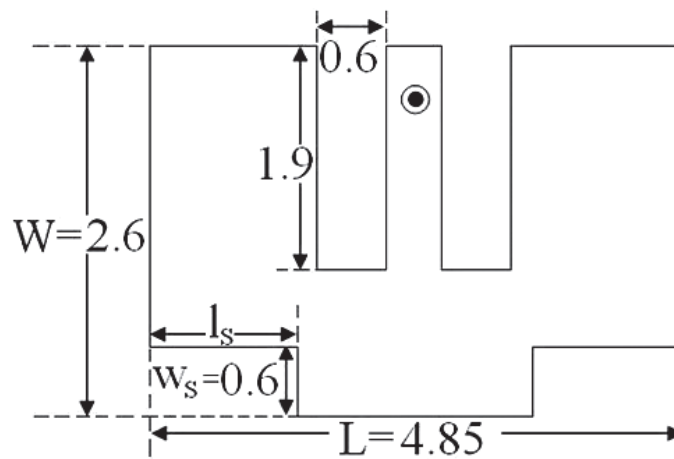


Figure 7c. The Ψ -shaped microstrip antenna.

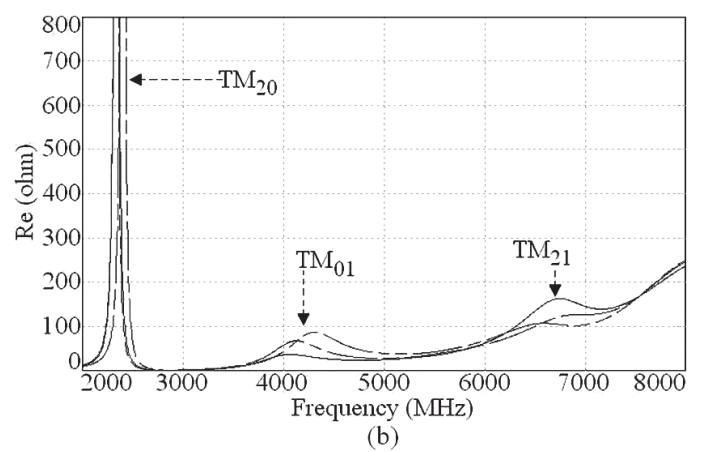
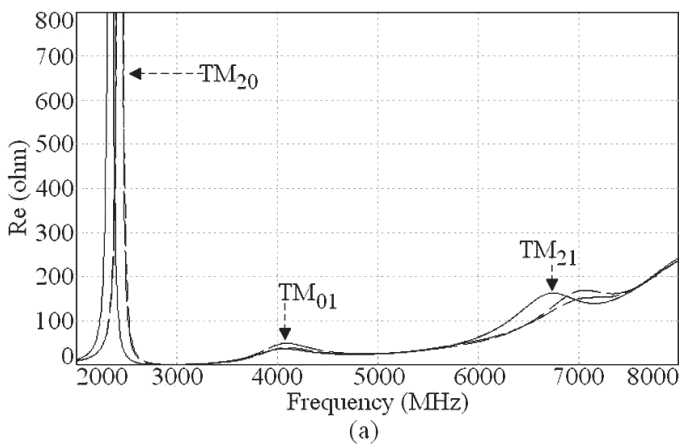


Figure 8. The resonance curve plots for (a) (—) $l_s = 0.0$ cm, (---) $l_s = 0.5$ cm, and (— · —) $l_s = 1.0$ cm; and (b) (—) $l_s = 0.0$ cm, (---) $l_s = 1.5$ cm, and (— · —) $l_s = 2.0$ cm, for the Ψ -shaped microstrip antenna.

slots. Since the surface currents varied along the patch's length as well as along its width, the radiation pattern showed higher cross-polarization levels.

Furthermore, to realize the Ψ -shaped microstrip antenna, a pair of rectangular slots was cut on the other radiating edge of the E-shaped microstrip antenna, as shown in Figure 7c. The slot length was increased in steps of 0.5 cm, and the resonance-curve plots for this antenna are shown in Figures 8a and 8b.

An additional pair of slots was cut along the patch's length. The pair of slots was parallel to the surface currents for the modified TM_{20} mode, and hence the reduction in its resonance frequency was negligible. For larger l_s , the additional pair of slots reduced the effective patch width, which increased the TM_{01} -mode frequency of the patch. The TM_{21} -mode resonance frequency increased slightly for smaller values of l_s , because with an increase in slot length, the effective patch dimensions were reduced. However, for $l_s > 1.5$ cm, the TM_{21} frequency was reduced because the surface currents were forced to travel along the slot's length. For $l_s = 2.3$ cm, along with the loop formed due to the E-shaped patch, a loop due to the TM_{21} mode was also optimized inside the $VSWR = 2$ circle in the impedance plot and a broadband response was obtained, as shown in Figure 2a. In this optimized configuration, the modified TM_{21} mode was dominant towards the higher frequencies, and the surface currents were directed along the horizontal as well as the vertical directions. Hence, the radiation patterns at those frequencies (> 6000 MHz) showed higher cross-polarization levels, which reduced the antenna's gain from more than 10 dBi to less than 4 dBi. For frequencies above 6000 MHz, the total substrate thickness was also more than $0.123\lambda_0$. This thicker substrate led to unwanted radiation from the feeding probe in the end-fire direction, which also contributed to the reduction in the antenna's gain. Using the proximity-feeding technique, an improvement in the gain characteristics over the bandwidth, as well as a further increase in the antenna's bandwidth, was obtained, as discussed in following section.

4. A Proximity Fed Ψ -Shaped Microstrip Antenna

The proximity fed Ψ -shaped microstrip antenna is shown in Figures 9a and 9b. The configuration was optimized using an air substrate. The Ψ -shaped patch was suspended in air with a thickness of 0.7 cm, such that the total substrate thickness was nearly the same as that used in the probe-fed configuration. The coupling strip was placed below the patch at a thickness of 0.6 cm. The dimensions of the Ψ -shaped patch were kept the same. To optimize the position of the two loops of the Ψ -shaped patch inside the $VSWR = 2$ circle, the strip position below the patch and its dimensions were optimized. The input-impedance locus for this is shown in Figure 9c. The simulated bandwidth was from 4093 MHz to 8259 MHz (4167 MHz, 67.5%). A measurement was carried out, and the measured

bandwidth was from 4123 MHz to 8310 MHz (4187 MHz, 67.4%). Compared to the probe-fed configuration, because the air substrate was used ($\epsilon_r = 1$), the center frequency of the band increased. The radiation pattern and gain over the bandwidth were measured. The radiation pattern was in the broadside direction, with the E and H planes aligned along $\Phi = 90^\circ$ and 0° , respectively. The gain was more than 7 dBi over the VSWR bandwidth, with a peak gain very close to 9 dBi, as shown in Figure 10a. Although this configuration was optimized on a slightly thicker air substrate and at a higher center frequency, the gain towards the higher frequencies of the bandwidth was only reduced by 2 dBi compared to the maximum value, since proximity feeding was used.

Furthermore, the broadband behavior of the proximity-fed Ψ -shaped microstrip antenna was investigated in the 1000 MHz frequency band. This frequency band was selected because the proposed antenna could find applications in the mobile communication environment in the 800 MHz to 1200 MHz frequency range. To realize a larger bandwidth, a thicker air substrate of $h = 3.0$ cm was selected. The rectangular microstrip antenna dimensions were selected such that it resonated at around 850 MHz for the TM_{01} mode, and this gave a patch width (W) equal to 13 cm. Similarly to the reported Ψ -shaped configuration, a higher aspect ratio was selected. The patch length (L) was taken equal to 24 cm, which gave $L/W = 1.8$. For this rectangular microstrip antenna dimension, the various resonant-mode frequencies in the 800 MHz to 1800 MHz frequency range were TM_{10} : 519 MHz; TM_{01} : 857 MHz; TM_{20} : 1036 MHz; TM_{02} : 1714 MHz; TM_{21} : 1359 MHz; and TM_{12} : 1791 MHz. The rectangular microstrip antenna was fed using a coupling strip that was placed below the patch, as shown in Figures 10b and 10c. The simulated resonance-curve plot for this configuration is shown in Figure 10d. Since the feed was placed along the line joining the center of the patch and the center of the length, it did not excite the TM_{10} mode. The first peak in the resonance curve was due to the TM_{01} mode. The next two peaks corresponded to the TM_{20} and TM_{21} modes.

Inside this rectangular microstrip antenna, a pair of slots was first cut to realize the E-shaped microstrip antenna shown in Figure 10b. The resonance-curve plots for a variation in w_e from 2.0 cm to 7.0 cm are shown in Figures 11a and 11b. The resonant frequencies of the TM_{20} and TM_{21} modes reduced with an increase in w_e , whereas the TM_{01} -mode frequency remained nearly unchanged. The optimum bandwidth was realized for $w_e = 10.0$ cm, when the loop formed due to the coupling between the modified TM_{20} and TM_{01} modes was inside the $VSWR = 2$ circle, as shown in Figure 12a. The simulated bandwidth was 300 MHz at a center frequency of 906 MHz. The input-impedance plot showed the formation of a second loop due to the modified TM_{21} mode, at a frequency of 1277 MHz.

Furthermore, an increase in the bandwidth was realized by optimizing the second loop's position inside the $VSWR = 2$

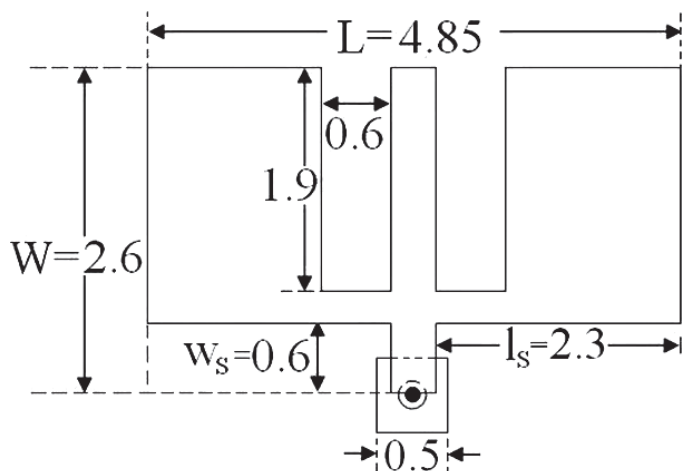


Figure 9a. A top view of the proximity-fed Ψ -shaped microstrip antenna.

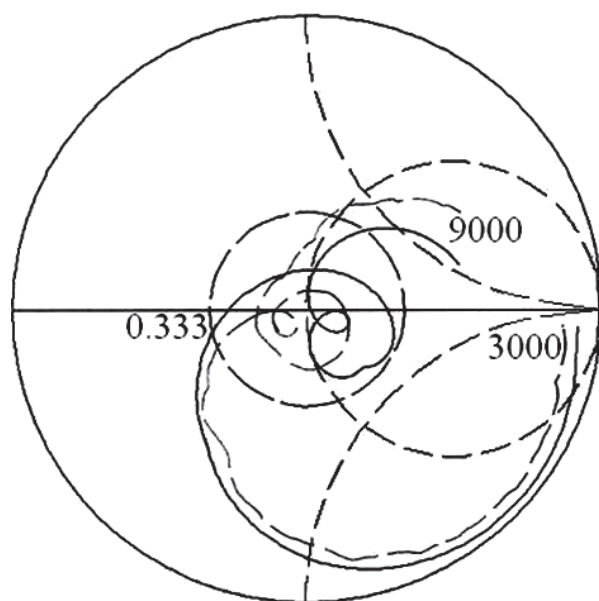


Figure 9c. The input impedance of the proximity-fed Ψ -shaped microstrip antenna: (—) simulated, (---) measured.



Figure 9b. A side view of the proximity-fed Ψ -shaped microstrip antenna.

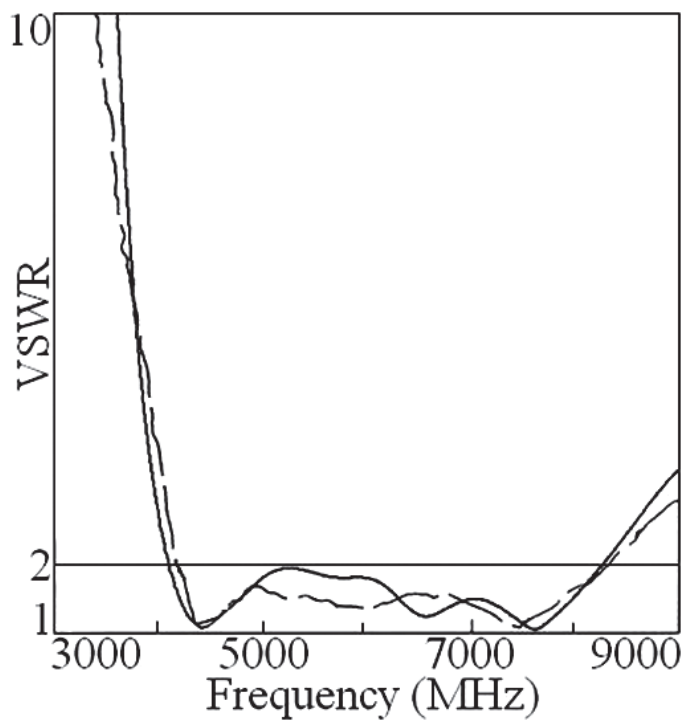


Figure 9d. The VSWR of the proximity-fed Ψ -shaped microstrip antenna: (—) simulated, (---) measured.

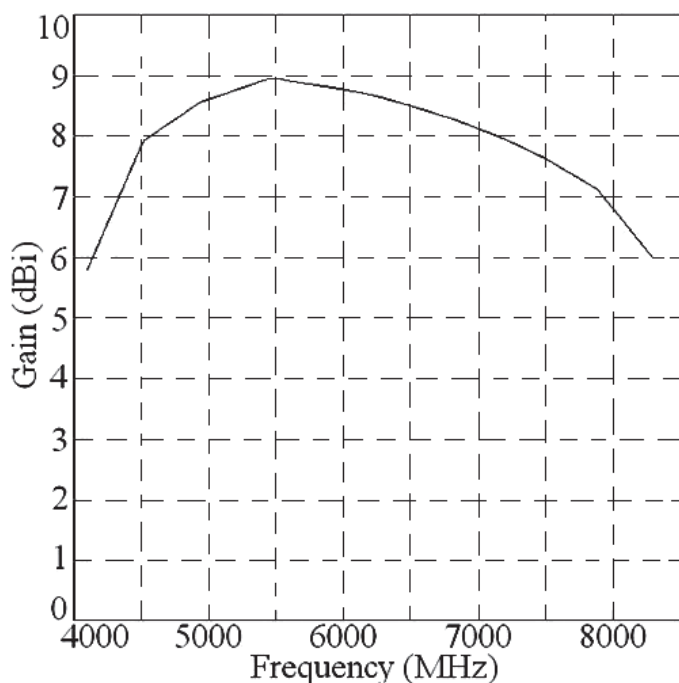


Figure 10a. The gain variation over the bandwidth for of the proximity-fed Ψ -shaped microstrip antenna.



Figure 10c. A side view of the proximity-fed rectangular microstrip antenna.

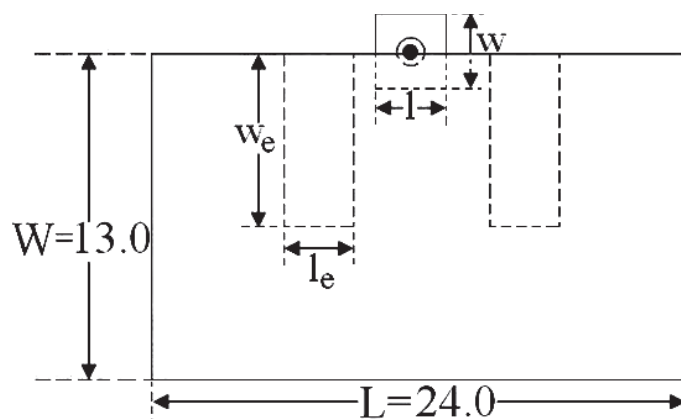


Figure 10b. A top view of the proximity-fed rectangular microstrip antenna.

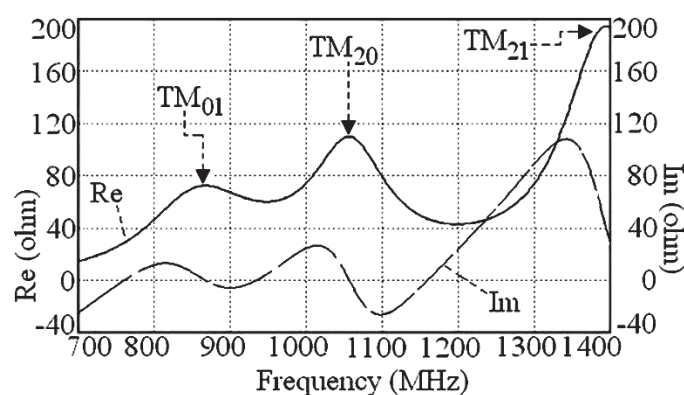


Figure 10d. The resonance curve plot for the proximity-fed rectangular microstrip antenna in the 1000 MHz band.

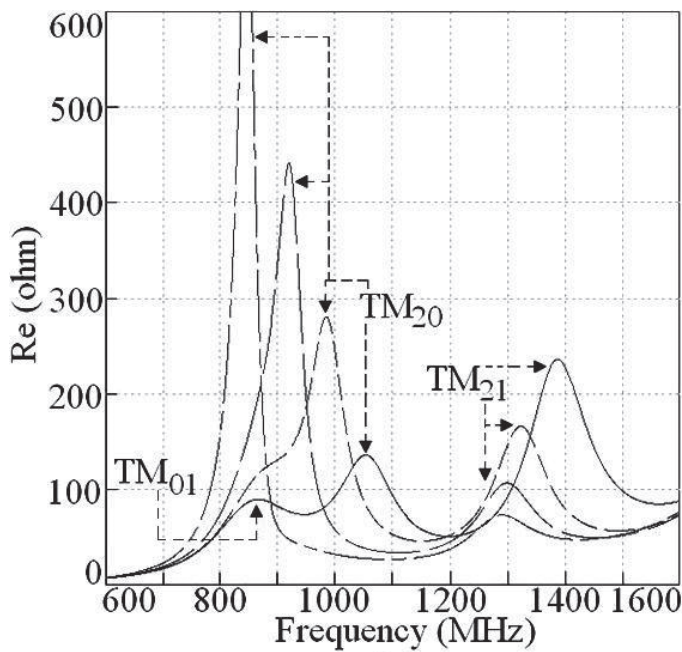


Figure 11a. The resonance curve plot for the E-shaped microstrip antenna for different slot lengths: (—) rectangular microstrip antenna for reference; (---) $w_e = 2.0$ cm, (— · —) $w_e = 3.0$ cm, (---) $w_e = 4.0$ cm.

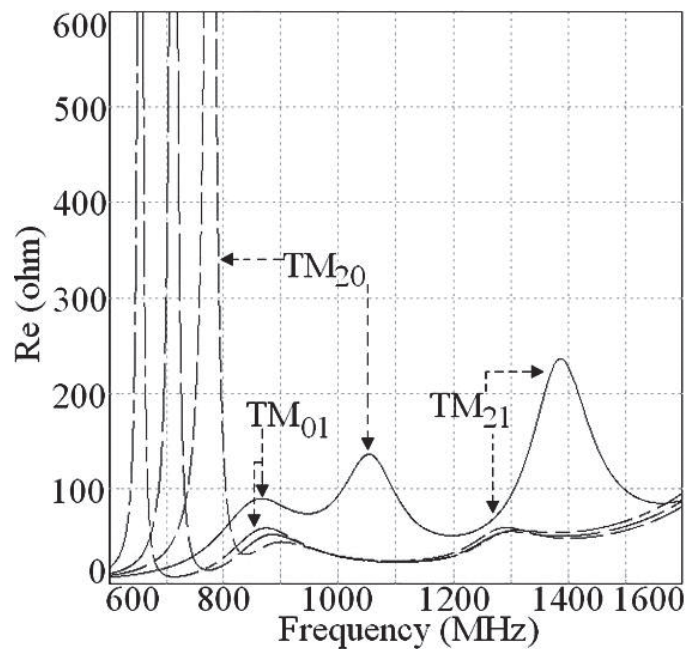


Figure 11b. The resonance curve plot for the E-shaped microstrip antenna for different slot lengths: (—) rectangular microstrip antenna for reference; (---) $w_e = 5.0$ cm, (— · —) $w_e = 6.0$ cm, and (---) $w_e = 7.0$ cm.

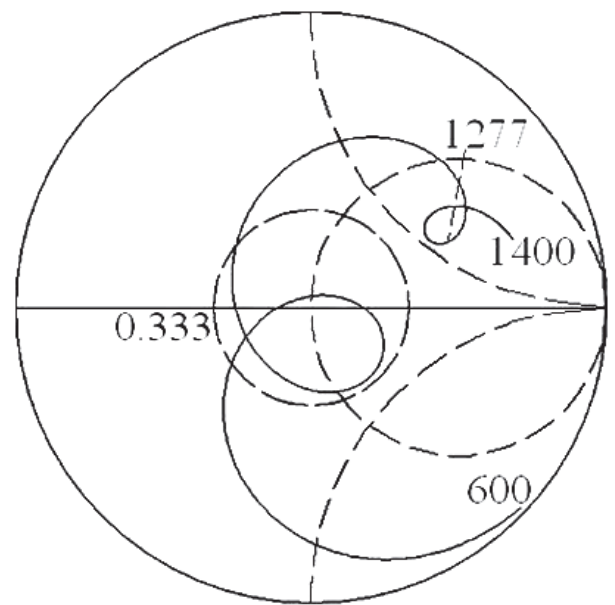


Figure 12a. The input-impedance plot for the proximity-fed E-shaped microstrip antenna at 1000 MHz.

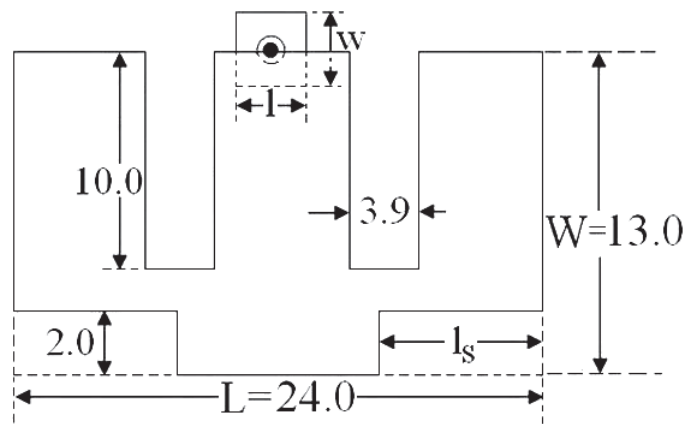


Figure 12b. The proximity-fed Ψ -shaped microstrip antenna at 1000 MHz.

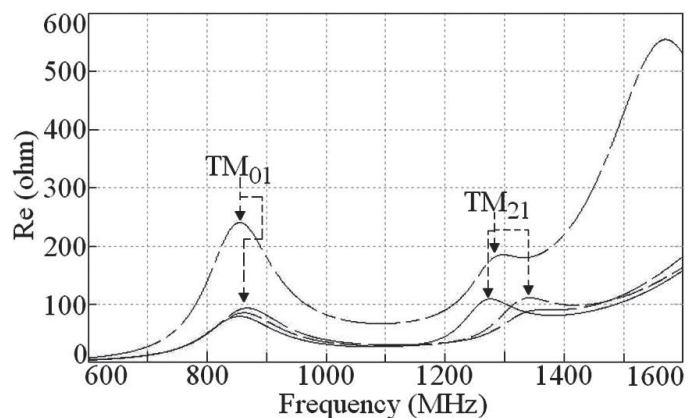


Figure 12c. The resonance curve plots for the proximity-fed Ψ -shaped microstrip antenna: (—) E-shaped microstrip antenna for reference; (---) $l_s = 2.0$ cm, (— · —) $l_s = 4.0$ cm, and (---) $l_s = 6.0$ cm.

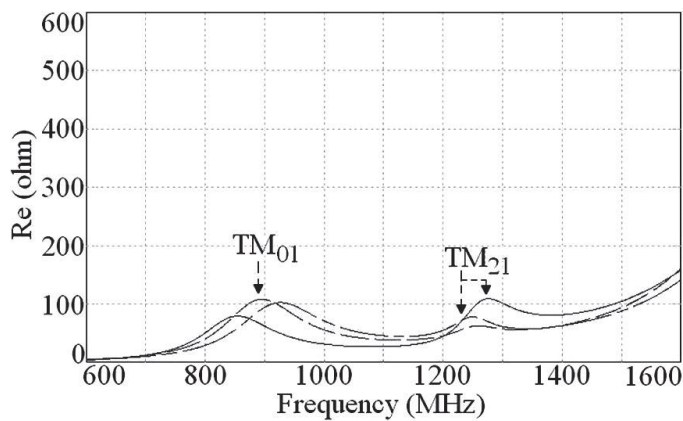


Figure 13a. Resonance curve plots for the Ψ -shaped microstrip antenna in the 1000 MHz frequency band: (—) E-shaped microstrip antenna for reference; (---) $l_s = 8.0$ cm and (— · —) $l_s = 10.0$ cm.

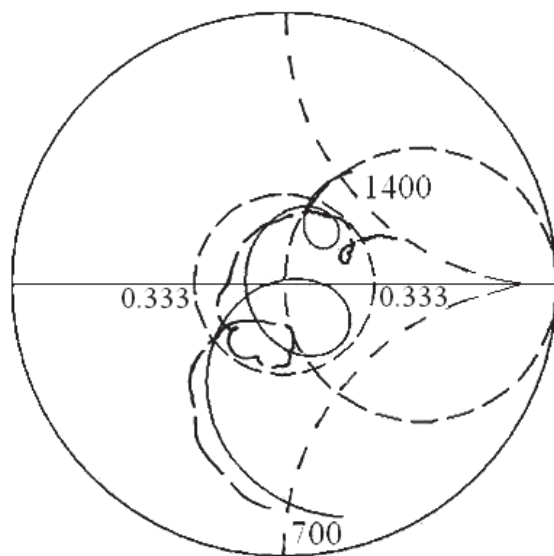


Figure 13b. The input impedance for the Ψ -shaped microstrip antenna in the 1000 MHz frequency band: (—) simulated and (---) measured.

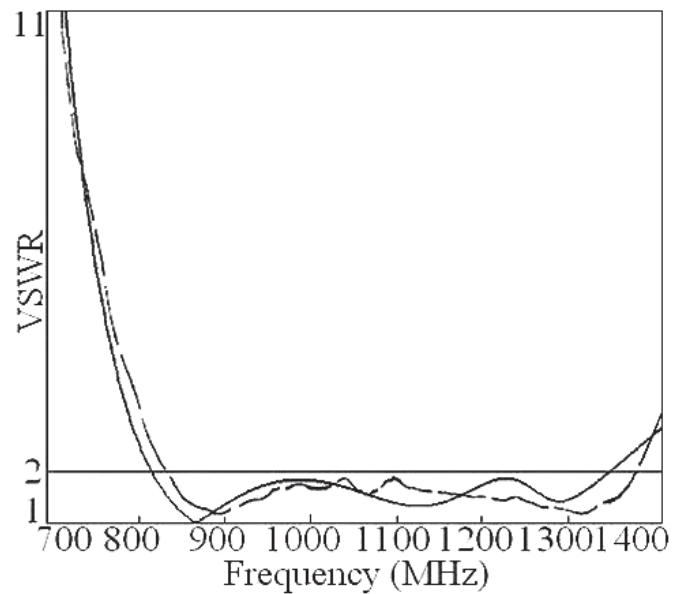


Figure 13c. The VSWR for the Ψ -shaped microstrip antenna in the 1000 MHz frequency band: (—) simulated and (---) measured.

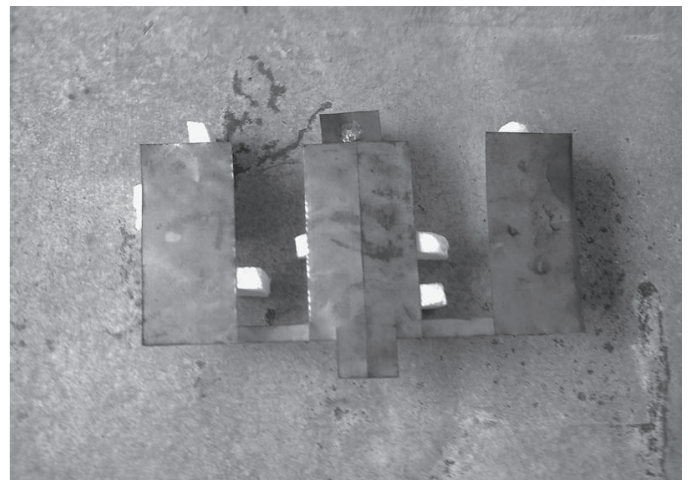


Figure 14. The fabricated prototype of the proximity-fed Ψ -shaped microstrip antenna in the 1000 MHz frequency band.

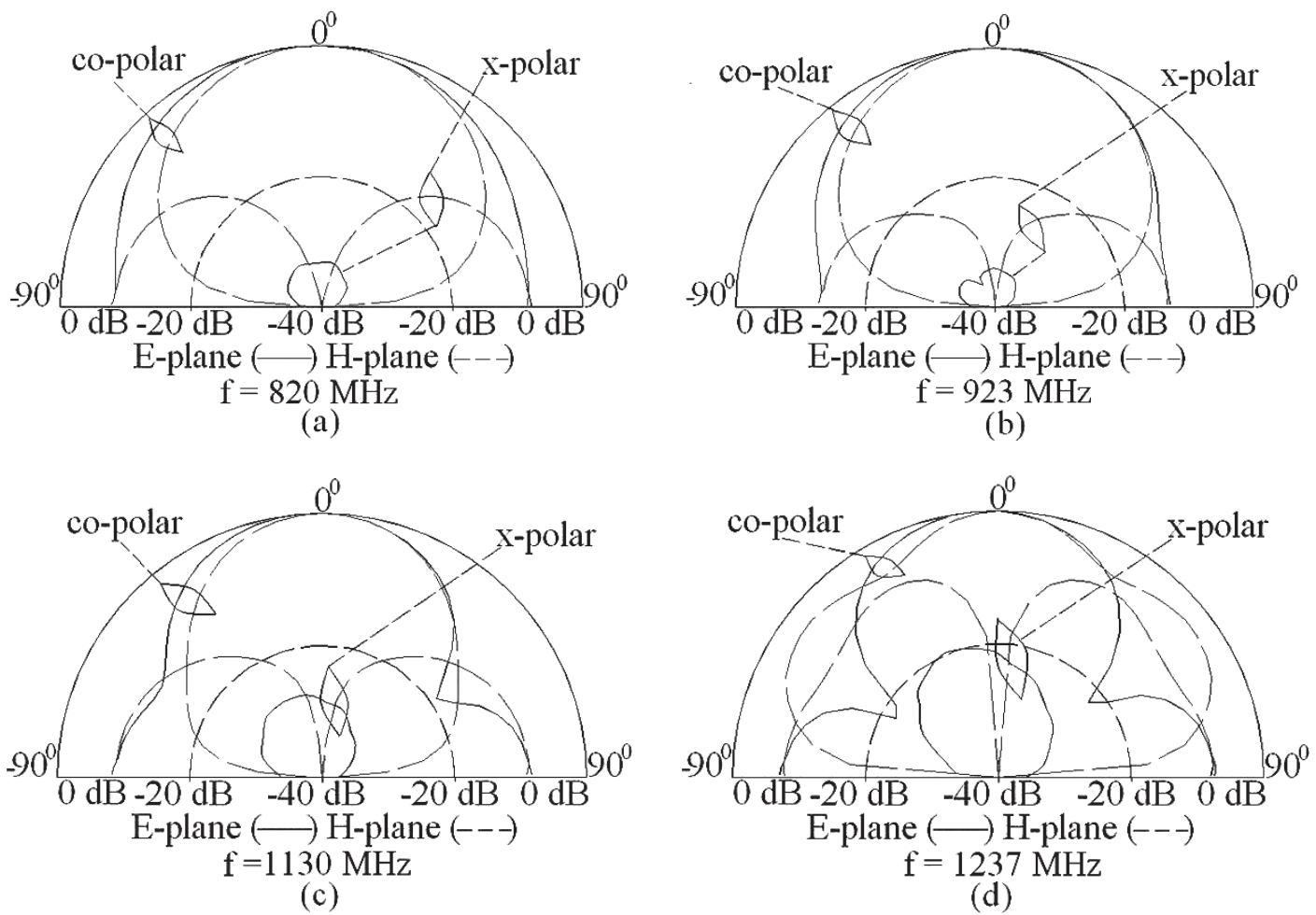


Figure 15a-15d. The radiation patterns over the bandwidth for the proximity-fed Ψ -shaped microstrip antenna in the 1000 MHz frequency band.

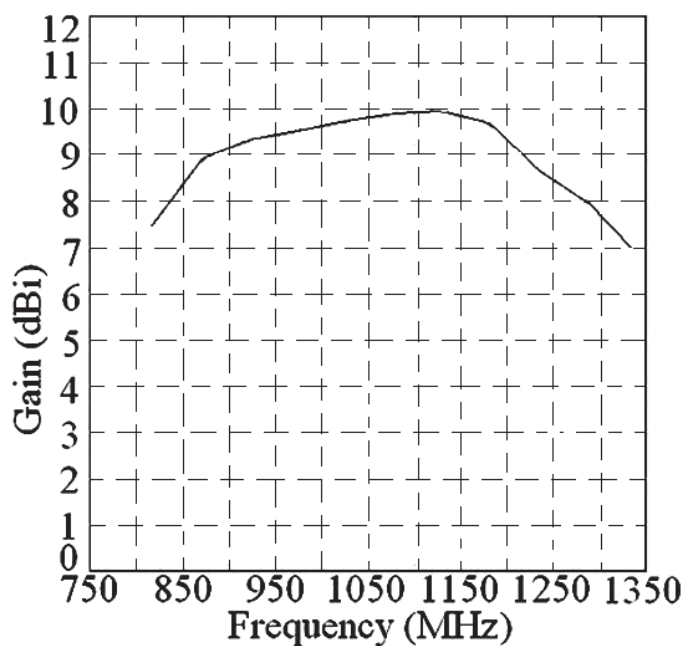


Figure 15e. The gain variation over the bandwidth for the proximity-fed Ψ -shaped microstrip antenna in the 1000 MHz frequency band.

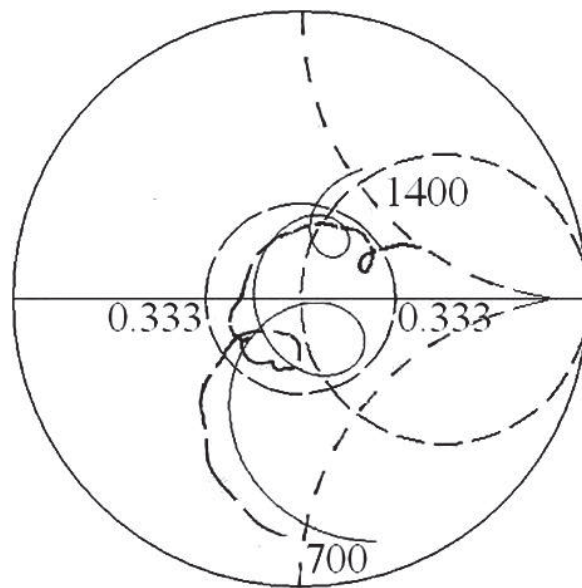
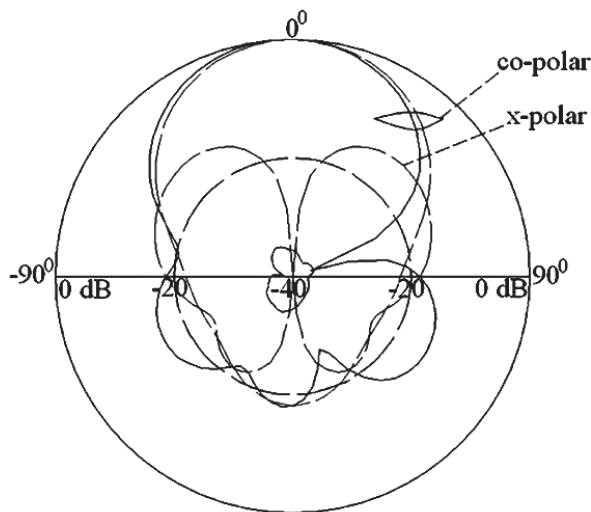
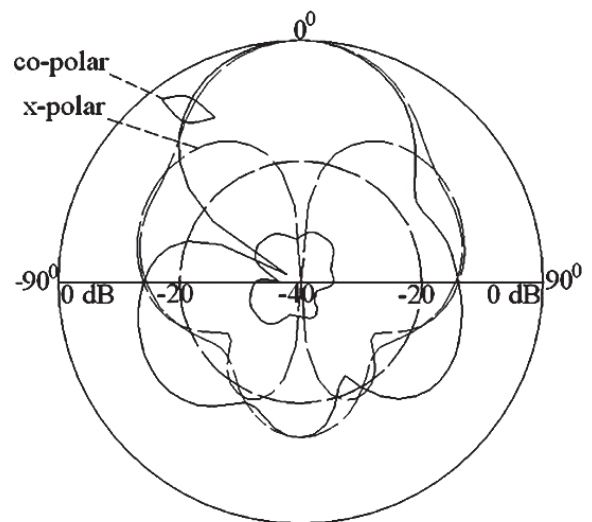


Figure 16a. The input-impedance locus over the bandwidth using a finite ground plane for the proximity-fed Ψ -shaped microstrip antenna in the 1000 MHz frequency band: (—) simulated and (---) measured.



(—) E-plane (---) H-plane
(b) $f = 1020$ MHz



(—) E-plane (---) H-plane
(c) $f = 1260$ MHz

Figures 16b, 16c. The radiation patterns over the bandwidth using a finite ground plane for the proximity-fed Ψ -shaped microstrip antenna in the 1000 MHz frequency band: (—) simulated and (---) measured.

circle. This was obtained by cutting an additional pair of slots that completed the Ψ -shaped structure, as shown in Figure 12b. The resonance-curve plots for varying values of l_s are shown in Figures 12c and 13a. Similarly to the above configuration, for smaller values of l_s (i.e., $l_s < 5$ cm), the TM_{21} mode frequency increased, whereas for larger values of l_s , its frequency was reduced. The TM_{01} -mode frequency increased for larger values of l_s (i.e., $l_s > 7$ cm). The optimum broadband response was realized for $l_s = 10.0$ cm, when both of the loops were inside the $VSWR = 2$ circle, as shown in Figure 13b. The simulated bandwidth was 517 MHz (47.5%).

The antenna was fabricated using a copper plate, and was supported in air using a foam substrate that was placed towards the antenna's corners. The measured bandwidth was 524 MHz (47.7%). The fabricated prototype of the proximity-fed Ψ -shaped microstrip antenna is shown in Figure 14. The radiation patterns over the bandwidth at four frequencies are shown in Figures 15a to 15d. The E and H planes were aligned along $\Phi = 90^\circ$ and 0° , respectively. The radiation pattern was in the broadside direction, with a cross-polarization level of less than 15 dB compared to the co-polar levels. Since the TM_{21} mode was dominant towards the higher frequencies of the bandwidth, the cross-polarization levels were higher. The gain variation over the bandwidth was measured, and it is shown in Figure 15e. The proposed antenna had a gain of more than 7 dBi over the entire bandwidth, with a peak gain nearly equal to 10 dBi. As the proximity-fed configuration was used, the decrease in gain was only 3 dBi. Since the simulation was carried out using an infinite ground plane, these antenna characteristics were measured using a larger square ground plane, with a side length of 80 cm ($2.67\lambda_0$). The backlobe radiation was thus absent.

The above antenna characteristics were also verified using a finite square ground plane with a side length of 28 cm ($0.93\lambda_0$). The measured and simulated input-impedance plots and radiation patterns at two frequencies over the bandwidth are shown in Figures 16a to 16c). Similar impedance characteristics were obtained using the finite ground plane, with simulated and measured bandwidths of 509 MHz (46.9%) and 521 MHz (47.4%), respectively. The radiation pattern was in the broadside direction, with the backlobe radiation being 15 dB down compared to the main-lobe boresight radiation. Towards the higher frequencies of the bandwidth, the backlobe radiation

was only 10 dB down compared to the main-lobe levels. The results of all of the configurations are given in Table 1.

5. Conclusions

The broadband response of the reported Ψ -shaped microstrip antenna was studied. The Ψ -shaped microstrip antenna was realized by cutting an additional pair of slots on the other radiating edges of the E-shaped microstrip antenna. The resonance-curve plots and surface-current distributions for the equivalent rectangular microstrip antenna, the E-shaped microstrip antenna, and the Ψ -shaped microstrip antenna were studied. It was observed that the pair of slots that was cut to realize the E-shaped microstrip antenna modified the resonance frequency of the higher-order TM_{20} mode and, along with the TM_{01} mode, produced a broader bandwidth. Since the slot modified the directions of the surface currents for the TM_{20} mode, the radiation pattern over the entire bandwidth remained in the broadside direction, with no variations in the directions of the principal planes. Furthermore, when an additional pair of slots was cut on the other edges of the E-shaped microstrip antenna, they modified the resonance frequency of the higher-order TM_{21} mode. Due to its coupling with the modified TM_{20} and TM_{01} modes, this resulted in a further increase in the bandwidth in the Ψ -shaped microstrip antenna. Since the TM_{21} mode was dominant towards the higher frequencies of the bandwidth, the cross-polar radiation increased, which reduced the gain.

A proximity-fed configuration of the Ψ -shaped microstrip antenna in the 5500 MHz frequency band was proposed. Due to the use of the proximity feeding, the reduction in the gain towards the higher frequencies of the bandwidth was only 2.5 dBi.

Proximity-fed configurations of the Ψ -shaped microstrip antenna in the 1000 MHz frequency band, using infinite as well as finite ground planes, were proposed. They yielded simulated and measured bandwidths of more than 500 MHz ($>45\%$), with a broadside radiation pattern and a maximum gain very close to 10 dBi. Improvement in the gain towards the higher frequencies of the bandwidth was obtained, with the reduction at the higher end being only 3 dBi.

Table 1. A comparison among various broadband Ψ -shaped microstrip antennas.

Configuration	Simulated BW	Measured BW
Ψ -shaped MSA (infinite ground plane)	3306 MHz, 60.1%	3280 MHz, 60%
Proximity-fed Ψ -shaped MSA	4167 MHz, 67.5%	4187 MHz, 67.4%
Proximity-fed Ψ -shaped MSA at 900 MHz	517 MHz, 47.5%	524 MHz, 47.7%
Proximity-fed Ψ -shaped MSA at 900 MHz, finite ground plane	509 MHz, 46.9%	521 MHz, 47.4%

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